

Box 2: Input and Output

User input

Frequency	2.4 GHz
Dielectric constant	4.8
Height	6 mm

Program output

Patch length	25.99 mm
Patch width	36.70 mm
Eff. dielectric constant	4
Eff. length	31.23 mm
Length extension	2.62 mm
Ground length	61.99 mm
Ground width	72.70 mm
Yf	18.35 mm
Xf	6.49 mm
Bandwidth in percentage	25.10%

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

It is compiled using Turbo C version 3.0. Here, C graphics files are used to provide a better, more user-friendly interface of the program. The program starts with the following lines:

```
#include<stdio.h>
#include<graphics.h>
#include<stdlib.h>
#include<ctype.h>
#include<dos.h>
#include<math.h>
```

The program computes various parameters of the patch antenna including its length and width, the effective dielectric constant, feed point and bandwidth.

The patch width is defined by the following line in the code:

```
patch_width=(c)/(2*frequency*sqrt
((dielectric_constant+1)/2));
```

The patch length is defined by the following line in the code:

```
patch_length=effective_length-2*
length_extension;
```

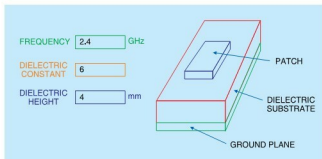


Fig. 2: User's input example

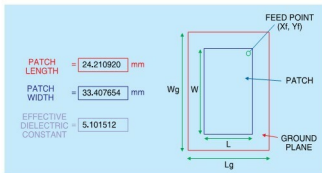


Fig. 3: Example's output

The bandwidth of the patch antenna is calculated using the following subroutine:

```
void bandwidth_calculation()
{ if (bandwidth_factor<=0.045)
A=180;
else if (bandwidth_factor>=0.045
&&bandwidth_factor<=0.075)
A=200;
else if (bandwidth_factor>=0.075)
A=220;
bandwidth=(A*dielectric_height*sqrt
(patch_width/patch_length))/
wavelength_in_substrate;
settextstyle(2,0,5);
setcolor(1+random(14));
int bandwidth_box_start_x=120,
bandwidth_box_start_y=110;
```

Running the software

1. Install the Turbo C software in your local disk's C drive.
2. Compile the code to get an ex-

ecutable file with .exe extension.

3. Run the .exe file.

4. On the very first screen, the program prompts you to give input parameters like the frequency in GHz, the dielectric constant, and the height of the dielectric constant in millimetres. A typical example for a user's input is shown in the screenshot Fig. 2.

5. Press any key to proceed. The program calculates length, width and other parameters like ef-

fective length and length extension, length and width of ground plane, position of feed points in terms of co-ordinates, etc. The length, width and effective dielectric constant of the patch antenna for the example is shown in Fig. 3.

6. Finally, the bandwidth as an improved percentage will be calculated.

Another example with both input and output is shown in Box 2. The resonant frequency selected for this design is again 2.4 GHz, and the dielectric constant is 4.8.

You can cross check the results with a similar, yet reliable software from the Internet, which you can find at <http://www.emtalk.com/mscalc.php> •

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